



Research Paper

Techno-Economic and life cycle assessment of standalone Single-Stream material recovery facilities in the United states

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ABSTRACT

Material Recovery Facilities (MRFs) are crucial players in achieving a circular economy. MRFs receive complex waste streams and separate valuable recyclables from these mixtures. This study conducts techno-economic analysis (TEA) to estimate the net present value (NPV) and life cycle assessment (LCA) to estimate different environmental impacts of a commercial scale standalone, single-stream MRF to assess the economic feasibility and environmental impacts of recovering valuable recyclables from an MRF processing 120,000 tonnes per year (t/y). The TEA employs a discounted cash flow rate of return (DCFROR) analysis over a 20-year facility lifetime, along with a sensitivity analysis on the impact of different operating and economic parameters. Results show that the total fixed cost of building the MRF facility is \$23 MM, and the operating cost is \$45.48/tonne. The NPV of the MRF can vary from \$3.57 MM to \$60 MM, while 100-year global warming potential can range from 5.98 to 8.53 kg carbon dioxide equivalents (CO₂-eq) per tonne of MSW. We have also found that MSW composition (arising from regional effects) significantly impacts costs, 100-year global warming potential, and other impact categories such as acidification potential, eutrophication potential, ecotoxicity, ozone depletion, photochemical oxidation, carcinogenic effects, and non-carcinogenic effects. Sensitivity and uncertainty analysis indicate that waste composition and market prices significantly impact the profitability of the MRF, and the waste composition mostly impacts global warming potential. Our analysis also indicates that facility capacity, fixed capital cost, and waste tipping fees are vital parameters that affect the economic viability of MRF operations.

1. Introduction

Achieving a circular economy is key to preventing depleting resources and environmental degradation (Khan & Haleem, 2021). A circular economy aims to achieve this through recycling and upcycling materials and energy at the consumer, producer, urban, regional, and national levels (Prieto-Sandoval et al., 2018). To achieve high degrees of circularity, it is necessary to develop strategies and technologies to recover and generate value from waste (Lase et al., 2023; Prieto-Sandoval et al., 2018). In doing so, Material Recovery Facilities (MRFs) play a pivotal role (Cimpan et al., 2015) because they separate waste (recyclable waste and/or municipal solid waste), which is a complex waste mixture, into various streams that have varying levels of economic value. Furthermore, MRFs are key players in reducing waste

going to landfills and avoiding the associated social and environmental impacts.

MRFs are sophisticated industrial facilities that separate marketable recyclables from different waste streams and resell the recovered materials to secondary recyclers. MRFs are classified into four primary types based on the input waste streams they process (Combs, 2011): single-stream, dual-stream, mixed-stream, and presorted MRFs. Mixed-stream MRFs receive and process all kinds of MSW. Single-stream MRFs only process recyclable materials collected in dedicated bins from households, and commercial organizations. Dual-stream and presorted MRFs receive and process waste streams that have been presorted and are almost ready to be repurposed. Mixed-stream MRFs operated in the early 1990s in the U.S. achieved poor material recovery rates due to the presence of a significant amount of organics and other non-valuable

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materials in the waste stream, ultimately leading to financial and operational struggles (Cimpan et al., 2015). In recent years, single-stream MRFs have become more prevalent than other kinds of MRFs in the U.S. because of its increased recovery rate and community participation in storing all accepted recyclable materials in one bin (Fitzgerald et al., 2012). These facilities allow consumers to dispose of all their recyclable waste in one “recyclable” container. However, existing single-stream MRFs still face critical technical and economic challenges such as waste collection and transportation, uncertainty in waste stream compositions, tipping fees, sorting efficiencies, sorting cost, low-quality recyclable materials, and volatility of post-consumer recovered material market prices (Chaudhari et al., 2021; Cimpan et al., 2016; Milios et al., 2018).

Few studies have modeled the operations of MRFs, and even fewer studies have investigated the economic feasibility and environmental impacts of recovering value from waste in MRFs. Chang et al. (2005) studied the optimal design for the sustainable development of an average MRF in the U.S. in 2004. Franchetti (2009) conducted a study on the economic and operational feasibility of an MRF for municipal waste recycling in Ohio, U.S. These MRFs relied heavily on manual sorting, which is outdated compared to recent MRFs' state-of-the-art technology, which sorts with automated machines.

Recent studies have modeled the operations of MRFs considering more automated separation and recovery equipment (Combs, 2011; Pressley et al., 2015). Combs (2011) analyzed material recovery processes to characterize the relationship between crucial MRF parameters, processing cost, and resulting emission. Combs (2011) reported costs of \$15.16, \$9.72, \$11.06, and \$7.77 to process 1 tonne (t) of waste in single-stream, dual-stream, mixed-stream, and presorted recyclables MRFs, respectively. The author highlighted that a $\pm 25\%$ change in equipment efficiencies and diesel cost could result in a -10% to 5% change in total cost. This same paper also reported a global warming potential (GWP) of 14.94, 14.94, 11.40, and 3.62 kg CO₂-eq /t for single-stream, dual-stream, mixed-stream, and presorted recyclables MRFs, respectively. Moreover, this study also showed through sensitivity and uncertainty analysis that a $\pm 25\%$ change in diesel combustion from rolling stock used to haul materials around the MRF results in a $\pm 20\%$ change in CO₂ emissions. In addition, the study reported that waste composition is the most influential factor in the profitability of MRFs (Combs, 2011). Pressley et al. (2015) built on Comb's (2011) work and developed a process model to evaluate the MRFs' cost and energy use. Their model allows for estimating the labor and energy requirements, cost, and mass transfers, given the MRF configuration and the type of waste to be processed. For example, this study reported that the electricity needs of MRFs can range from 4.7 to 7.8. kWh/t of waste processed, and the amortized cost can vary from \$19.8 to \$24.9/t of waste processed (Pressley et al., 2015). The model developed by the authors is flexible and can be modified to analyze the four primary types of MRFs (Pressley et al., 2015). Although the study presented models to estimate the cost and emissions associated with a specific MRF configuration, the study did not consider market prices of recyclables to evaluate their return on investment (Pressley et al., 2015).

Cimpan et al. (2016) conducted a detailed techno-economic assessment (TEA) of central sorting at an MRF in Germany using lightweight packaging waste as a feedstock. The analysis considered four different plants ranging from 25,000 to 100,000 t/y and reported that capital investment could range between \$7.55 to \$22.64 MM. The operating costs increased by 240% when the scale increased by 300% (25,000 to 100,000 t/y). The study identified the feedstock's complex nature and the MRF's challenging processing conditions as significant factors that affect the recovery efficiencies and economic feasibilities of these plants (Cimpan et al., 2016).

Other studies have conducted life cycle assessments (LCA) on standalone MRFs to estimate the environmental impacts associated with such facilities. Fitzgerald (2012) analyzed and compared the GHG emissions of dual-stream and single-stream MRFs. The study assessed

the GHG emissions benefit or penalty associated with transitioning from dual-stream to single-stream MRFs. The author reported that single-stream MRFs could recover 50% more recyclables than dual-stream MRFs. Single-stream MRFs have significantly lower emissions than dual-stream MRFs, and for both MRF types, avoided emissions from recyclables are significantly higher than emissions of MRF operations. However, this study did not report the impact of waste composition on emissions. Smith et al. (2022) evaluated the total emissions associated with an MRF as part of an analysis focusing on recycling polyethylene terephthalate (PET). The study reported that MRFs emit 8.58 kg CO₂-eq /t of waste processed in the MRF. The study allocated 0.429 kg CO₂-eq of the total GHG emissions to separate PET from the waste stream (Smith et al., 2022). Based on the literature review, no study has conducted TEA and LCA of a standalone single-stream MRF that considers the impacts of waste composition and market prices on net revenues while including multiple mid-point environmental impact categories. By addressing this gap, we aim to identify opportunities to improve the profitability and environmental sustainability of material recovery in MRFs.

This study investigates the economic feasibility and environmental impacts of a standalone MRF. We build upon the Pressley et al. (2015) process model that simulates a single-stream MRF and utilizes the mass and energy balance from the process model to conduct a TEA and LCA. This study also investigates the effect of waste composition sampled from four U.S. regions on the economics and environmental impacts associated with the single-stream MRF.

2. Methodology

The TEA and LCA performance of a single-stream MRF that processes 120,000 t/y of waste obtained from four different U.S. locations are investigated. The analysis employs a process model that provides mass and energy flow data of a commercial-scale MRF. The MRF receives waste streams of various compositions and separates them into various post-consumer recovered materials (PCRMs). Based on the waste stream, the net present value (NPV) and mid-point category environmental impacts of recovering recyclables are estimated. The analysis employs sensitivity and uncertainty analysis to evaluate the effect of critical parameters, such as the waste stream's composition, capital cost, and tipping fees, on the economic and environmental impacts of recovering the recyclables.

2.1. Waste composition

The baseline analysis employs the U.S. national average waste composition reported in the U.S. EPA Waste Characterization Report (US EPA, 2010). The U.S. EPA waste composition was selected because it includes all the materials that can be recovered in the MRF. The materials include various grades of plastics (high-density polyethylene (HDPE), PET, plastic film, and mixed plastics), iron, aluminum, glass, old-corrugated containers (OCC), and non-OCC. OCC is used in boxes, containers, or other packaging materials consisting of three or more layers of kraft paper. Non-OCC materials are all other fiber materials that are not classified as OCC materials. PET is typically used in packaging food and beverages, while HDPE is used in packaging bottles, milk jugs, bags, containers, and toys. Plastic films include trash bags, plastic labels, videotape, grocery bags, storage bags, sheet film plastic, etc. Most MRFs do not accept plastic films because it clogs process equipment. Our analysis assumes that the MRF we model has sufficient capacity to accept the plastic film. Mixed plastics consist of all other plastic grades that are not PET, HDPE, and plastic films.

The U.S. EPA waste stream recorded 40% of the recyclable waste stream as OCC. The TEA and LCA results might be overestimated by the amount of OCC present in the waste stream as OCCs are mostly baled in commercial locations, i.e., most of the OCC may not make it to an MRF that receives recyclables from curbsides (Pressley et al., 2015). To account for the large amount of OCC in the U.S. EPA stream and waste

stream composition variation regionally and over time, the study assesses three additional scenarios based on waste stream compositions that represent the spectrum of conditions observed in the U.S. The other waste compositions include Pennsylvania (with information from the Pennsylvania Recovered Composition Study (Beck, 2005), Oregon (comingled recyclables before and after processing with information from the Solid Waste Policy, and Program Development Section Land Quality Division Oregon Department of Environmental Quality (Spindel, 2011)), and Seattle (with information from the Seattle Public Utilities Residential Waste Stream Composition Study (Cascadia, 2007)). Going forward, the Oregon, Pennsylvania, and Seattle waste stream will be referred to as Oregon, Pennsylvania, and Seattle, respectively. The waste stream composition from different locations has been widely studied and confirmed to vary from state to state and municipality to municipality (Jhun et al., 2021; US EPA, 2010). The variation is due to the point the waste was characterized, type of community, level of industrialization, lifestyle, and waste management regulations (Abdel-Shafy & Mansour, 2018; Jhun et al., 2021; US EPA, 2010). Pennsylvania's waste stream recorded a low amount of OCC because the waste stream was characterized at the MRF (Beck, 2005). As stated earlier, we believe most OCCs in Pennsylvania could have been baled before the stream arrived at the MRF. Fig. 1 shows the percentage composition of each recyclable material in the various waste streams. The small or zero amount of plastic films in Seattle, Oregon, and Pennsylvania could be because the waste streams were characterized at recovery facility gates. Most recovery facilities do not accept plastic films in single-stream waste.

2.2. MRF process model

This study assumes that garbage trucks collect waste from curbside recycling bins and transport it to the MRF. The trucks deliver the waste to the MRF tipping area, which can accommodate up to three days of stock, ensuring a steady supply of waste.

Fig. 2 shows the process flow of the MRF. The recovery process begins with floor sorting, where sorters remove large materials from the waste stream. Rolling stock pushes the material to a drum feeder that distributes the commodities to a belt conveyor at a constant rate. The belt conveyor moves the waste to a presort location where sorters identify and remove large and bulky recyclables that can clog, jam, or damage the recovery equipment. Sorters remove the plastic films in the stream and direct the plastic films to vacuum, and recovered plastic films are sent to the two-way baler, where it is packaged. The preliminary results of a national-level MRF survey show that approximately 8 out of 58 respondent MRFs keep track of the plastic films they sort in

their facility, with most MRFs sending the plastic films to landfills (Personal communication with survey team, 2022). The rest of the waste stream goes through a series of screens, where OCC and non-OCC are recovered. The rest of the materials in the stream (mostly packaging containers) fall through and go to the glass breaker screen, where the glass in the stream is broken into cutlets and recovered with other fines in the stream. The mix of broken glass and fines goes through an air knife which separates the fines and other light materials from the glass. The glass is further separated based on color by an optical sorter. The separated glass is subjected to manual sorting to remove ceramics in the mix. The rest of the waste is then sent to optical sorters that recover the PET and another optical sorter that separates high-density polyethylene (HDPE) based on color. The remaining materials in the stream are sent to a magnet and eddy current separator to recover the ferrous and aluminum materials, respectively. The remaining stream is sent for manual sorting, where pickers recover other valuable commodities in the stream. The residuals, which represent a mix of all the materials that cannot be separated, are sent to a landfill.

The material and energy balance of the MRF is modeled based on the design described in Pressley et al. (2015). The mass of the recovered material is calculated by multiplying the mass of the waste material by the recovery equipment's accuracy, shown in table S1 in the SI. The model estimates the electricity consumption of each piece of equipment based on its motor rating and maximum throughput using Eq (1).

$$E_j = \frac{e_j^{\text{MaxMotor}} * f_j^{\text{MC}}}{m_j^{\text{MTP}} * f_j^{\text{MTP}}} \quad (1)$$

Where:

E_j is the electricity consumption of equipment j in kWh/t of waste input.

e_j^{MaxMotor} is the rated motor capacity,

f_j^{MC} is the fraction of motor capacity utilized,

m_j^{MTP} is the maximum throughput of equipment j , and,

f_j^{MTP} is the fraction of the equipment capacity used.

The electricity model assumes a linear relationship between the mass of the waste stream processed through each piece of equipment and the electricity consumption. The limitation of this assumption is that it generalizes that all equipment's efficiency is constant across all loads; electricity consumption is directly proportional to load. However, some studies have shown that motor efficiencies vary with load (Burt et al., 2008; Li et al., 2015). Therefore, some equipment's electricity usage does not have a linear relationship with the mass of materials they process.

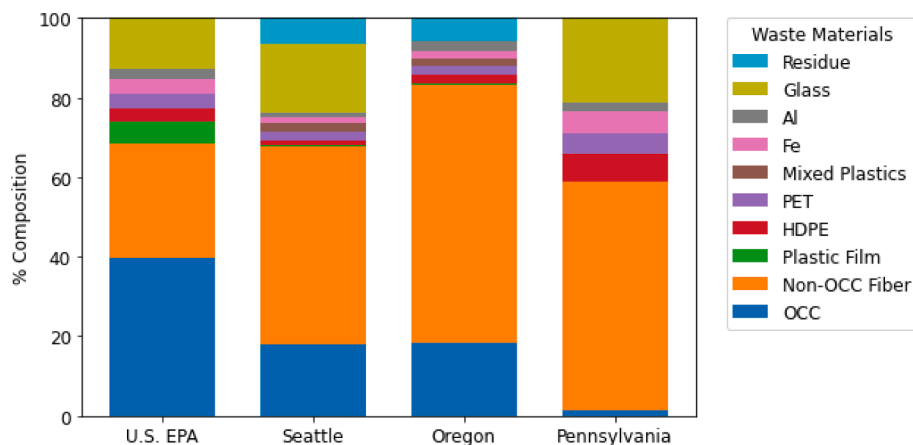


Fig. 1. Composition in weight % of recyclables and non-recyclables found in waste streams across different locations (EPA- Environmental protection agency, CC- Organic corrugated cardboard, Non-OCC- Non -organic corrugated cardboard, HDPE- High-density polyethylene, PET- polyethylene terephthalate, Fe – Ferrous metals, Al - Aluminum).

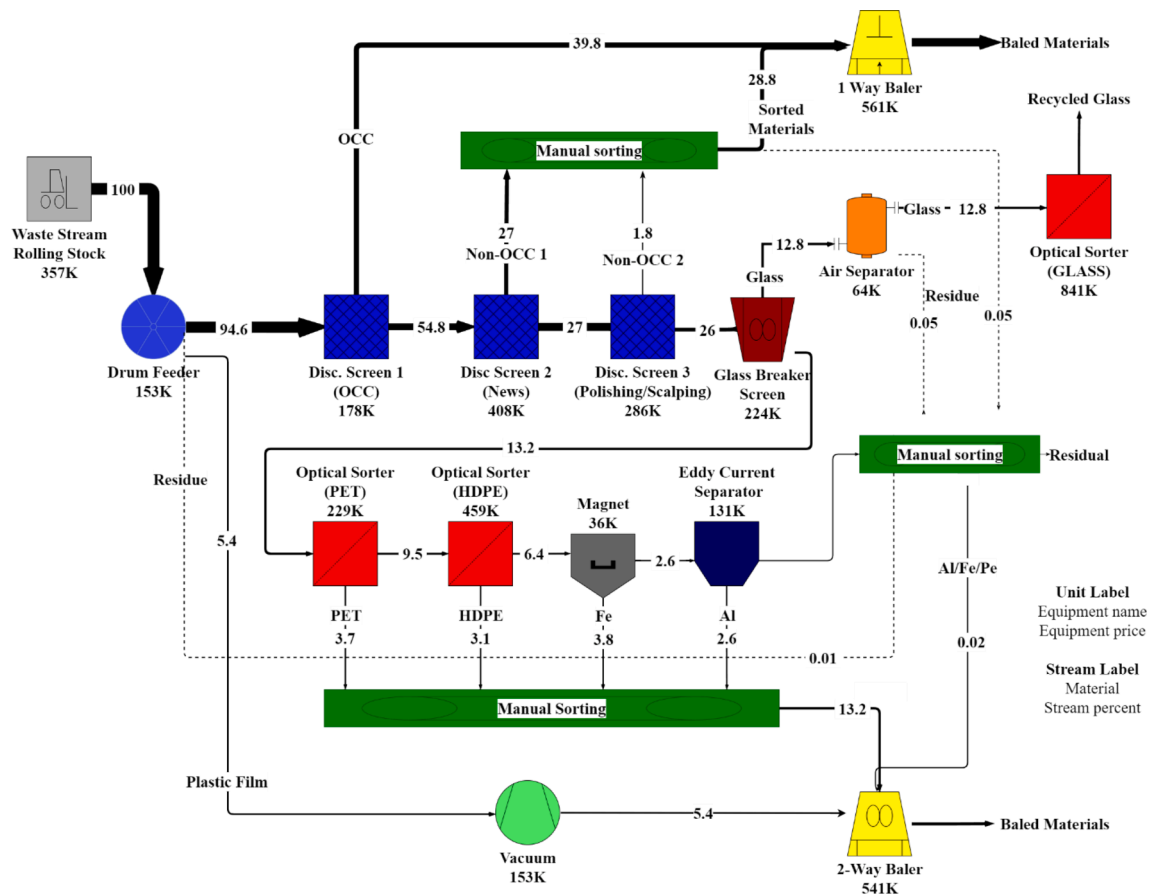


Fig. 2. Materials recovery facility (MRF) simplified process flow diagram with labels and cost of equipment. Weight and label of each stream represents the mass of material processed based on the U.S. EPA waste composition. (OCC– Organic corrugated cardboard, Non-OCC– Non-organic corrugated cardboard, HDPE- High-density polyethylene, PET- Polyethylene terephthalate, Fe – Ferrous metals, Al – Aluminum). (adapted and modified from Pressley et al. (Pressley et al., 2015)).

2.3. Techno-economic analysis

In this work, we use NPV as our primary economic indicator, which is estimated based on discounted cash flows of the MRF throughout its lifetime. We derive NPV as the difference between the present value of cash inflow and the facility's cash outflow over its lifetime (Žižlavský, 2014). NPV is a common metric used in evaluating the profitability of an investment. This study employs NPV because it facilitates comparisons across multiple products, unlike the minimum product selling price (MPSP), and over a broader range of profitability measures than the return on investment (ROI), which is challenging to interpret when the NPV is negative.

2.3.1. Financial assumptions

The TEA employs n^{th} plant assumptions (Meyer et al., 2020; Olafasakin et al., 2021). The n^{th} plant assumption assumes that the technology is mature, well-understood, and similar facilities have been built and are operating. It also implies that all technological breakthroughs needed for reliable and seamless facility operations have been accomplished. This is a realistic assumption as more than 335 MRFs are operating in the U.S. with similar equipment used in this analysis (Partnership, 2022). The study assumes that the cost of the MRF is financed through 40% equity and a 60% loan with a ten-year debt financing term at a 10% interest rate. The TEA estimates the NPV of the MRF, assuming a 10% internal rate of return (IRR) over 20-year plant life. The planning and construction period is assumed to be three years, and the start-up time is six months. All the financial assumptions we made for conducting the TEA are shown in Table 1.

Table 1

Financial assumptions for a 120,000 tonne per year materials recovery facility. MACRS (Modified Accelerated Cost Recovery System).

Parameters	Financial assumptions
Equity	40%
Financing interest rate	10%
Term for debt financing	10 years
Income tax rate	39%
Internal rate of return	10%
Plant-life	20 years
Construction period	3 years
Depreciation period	7 years MACRS schedule
Working capital	20% of Fixed capital cost
Project contingency	10% of Fixed capital cost
Plant salvage value	\$0
Start-up time	0.5 years
Revenue and cost during start-up	50% of normal revenue 75% of normal variable cost 100% of fixed cost
Electricity price	6.16 cents/kWh

2.3.1.1. Capital and operating cost. The capital cost includes the equipment purchase, installation, controls, piping, and electrical systems needed to integrate the equipment fully into the facility. The operating and maintenance cost (O&M) results from the routine maintenance and the replacement of equipment's consumable parts. Note that diesel and electricity costs are not included in O&M. To account for inflation, the equipment investment cost was adjusted to 2020 rolling dollars. Eq. (2) describes the estimation of the capital expenditure of equipment in the current year.

$$CAPEX = CAPEX_b \cdot \frac{CEPCI_c}{CEPCI_b} \quad (2)$$

CEPCI_c is the current year's chemical engineering plant-cost index,

CEPCI_b is the base year's chemical engineering plant-cost index,

CAPEX is the capital expenditure of equipment in the current year,

CAPEX_b is the capital expenditure of equipment in the base case.

The total fixed capital investment (TFCI) is the sum of the total direct and indirect costs. Total direct costs include investment and land costs, while indirect costs include engineering design, construction, project contingency, and legal and contractor fees. The construction cost was estimated by multiplying the MRF floor space by the construction cost rate of \$97.7 per square foot (Pressley et al., 2015). The MRF floor space required for processing 120,000 t/y is obtained from Combs (2011). The engineering design and contingency costs were assumed to be 32% and 37% of the investment cost, respectively (Pressley et al., 2015).

Operating costs are divided into fixed and variable costs. The fixed operating cost includes the maintenance of the equipment, labor, overhead, insurance, and depreciation, which are evaluated as percentages of total capital investment. Meanwhile, the overhead cost is estimated as a fraction of the labor cost. The detailed cost parameters are cited from Combs (2011) and are presented in Table S1 in SI. Information on the labor requirement of the MRF is supplied in section 2.1.2 of the SI.

Variable operating costs include the cost of waste collection and transportation, diesel for operating forklifts and rolling stock, electricity for offices, floor spaces, and running equipment, residue and waste disposal cost, income tax, and baling recyclables. Both electricity consumption and baling wire usage depend on the waste composition. Table S3 in the supporting information (SI) shows the summary of the operating cost assumptions.

To evaluate the facility's revenue, we use the PCRM market data for Chicago (Midwest/Central America) from recyclingmarkets.net (RecyclingMarkets, 2020). Figure S1 in the SI shows the market prices of PCRM as of Dec. 28, 2021. The prices of PCRM are very volatile (RBAC,

2019). They vary over time because of factors like oil prices, commodity demand, and market supply. The variation in prices over time does not follow a particular trend, and materials that cost more in the past can cost less in the future. To account for the high variation of post-consumer recovered material prices over time, a set of sensitivity analyses is conducted to evaluate the impacts of the variable market prices of products on the NPV.

2.4. Life cycle assessment

LCA is a practical and widely accepted methodology for estimating the health and environmental impact of a product system from raw material extraction through production, consumption, and disposal/recycling of the final product(s) (Finkbeiner et al., 2006). This study uses the LCA method, following the ISO 14040 and ISO 14044 framework (Finkbeiner et al., 2006), to assess the health and environmental impacts of processing one t of waste as delivered to the MRF, which is defined as the functional unit. The system boundaries are from cradle-to-gate. The system includes all activities within the MRF, and impacts related to upstream energy production and material inputs used at the MRF (Fig. 3). The study assumed a zero-burden approach for the waste received at the MRF, meaning that the environmental burdens associated with the production and use of the recovered waste materials before collection and transportation to the MRF are excluded. The zero-burden approach is a standard practice in solid waste management LCA (Voss et al., 2021). The cradle-to-gate scope covers all processes such as waste collection and transportation, equipment and office electricity consumption, rolling stock and conveyor diesel consumption, and waste residue disposal.

For the inventory analysis, four major hierarchy groups are modeled: waste collection and transportation to the MRF, recyclables separation, recovery, and packaging at the MRF, residue disposal, and avoided emissions. The energy and material balance from the process model serves as input data for the LCA. Residue disposal involves transporting residues to landfills. The study assumes that recovered materials in the

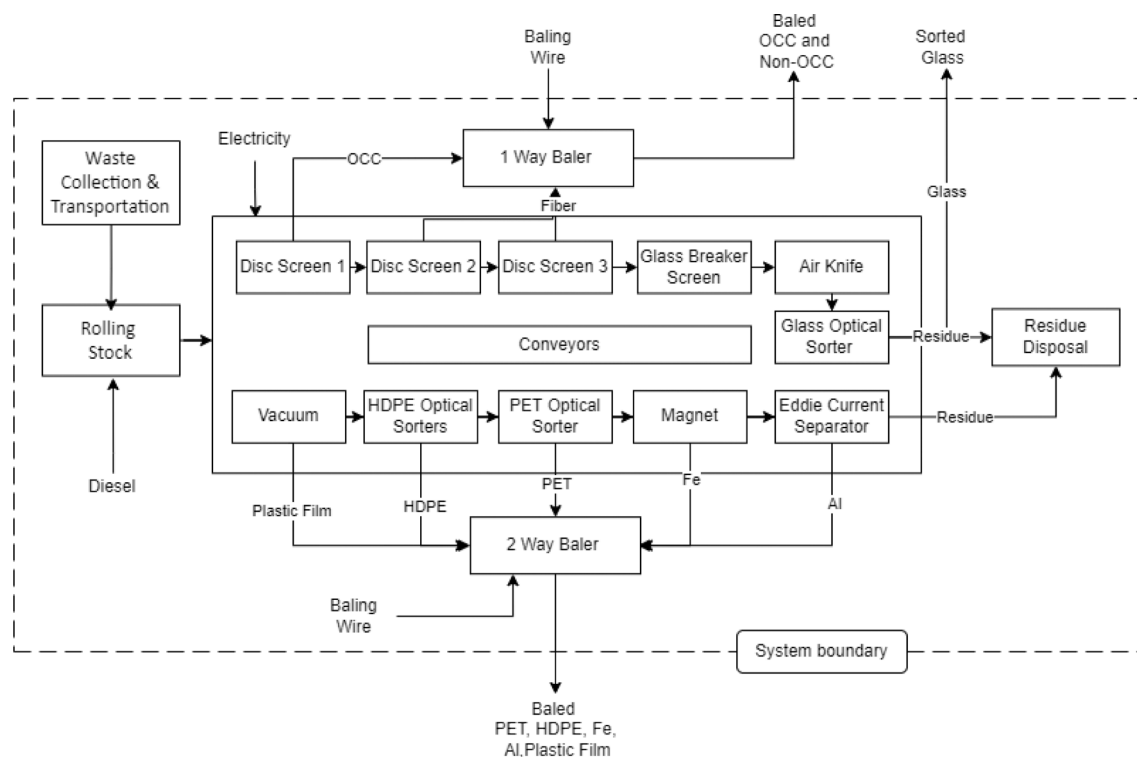


Fig. 3. System boundaries of processing recyclable waste in a single-stream material recovery facility, where OCC: Organic corrugated cardboard, Non-OCC: Non-organic corrugated cardboard, HDPE: High-density polyethylene, PET: Polyethylene terephthalate, Fe: Ferrous metals, Al: Aluminum).

MRF will be repurposed and used in place of virgin materials. Avoided emissions account for the impacts of displacing materials by MRF products in the market. This LCA approach is also known as the system expansion or substitution method. The study employs the OpenLCA (OpenLCA, 2022) software to analyze the inventory data using the United States Life Cycle Inventory Database (USLCI) (NREL, 2012). The study did not include the embodied emissions and decommissioning of the facility. McKinsey and Company reported that 24% of emissions of a process come from embodied emissions in the building, while 76% come from energy used for cooling and heating. In an industrial facility, where electricity for heating and cooling represents a small amount of the total energy needed, embodied emissions are significantly smaller than 24%. They can be negligible compared to total emissions. Table S5 in the SI shows the life cycle inventory data as collected from OpenLCA.

Environmental and health impacts can be evaluated at the midpoint or endpoint level using various life cycle impact assessment (LCIA) methods. Commonly used LCIA methods include ReCiPe, CML 2002, and eco-indicators (Al-Salem et al., 2014; Chen et al., 2019; Gu et al., 2017). This analysis employs the Tool for Reduction and Assessment of Chemicals and Other Environmental Impacts (TRACI) impact assessment method at the midpoint impact assessment level. The TRACI method is representative of U.S. conditions, and it groups the impact categories into environmental and human health impacts. The evaluated environmental impacts with TRACI include 100-year global warming potential (GWP) in kg CO₂-eq, acidification potential (AP) in H⁺-eq, eutrophication potential (EP) in kg N, ecotoxicity (ETP) in kg 2,4-D-eq, ozone depletion (ODP) in CFC-11-eq, and photochemical oxidation (PO) in kg NO_x-eq. The human health impacts include carcinogenic (Ca) in kg benzene eq and non-carcinogenic (NCa) in kg toluene eq. Table S4 in SI shows the emission factors used in this study.

The study assumed the average U.S. electricity mix in the LCA to highlight the impact of waste stream composition on the environmental impacts. However, it is important to highlight that the electricity mix varies from state to state and can significantly impact the environmental impacts of the MRF. Hence, the study assessed the effects of the varying electricity mix by location on the GWP, assuming the MRF is located in the state where the waste was collected. Emission factors for the individual state are obtained from the U.S. Energy Information Administration (EIA) electricity profiles (U.S. EIA, 2022).

2.5. Sensitivity and uncertainty analysis

Sensitivity analysis is a methodology employed to assess the variability of a model's output to critical model input parameters when each parameter is altered, and other variables are kept constant. Sensitivity analysis is used in this study to identify the parameters (changed by $\pm 30\%$) that most influence NPV and LCA impact assessment categories. For NPV, the changed variables are operating hours, waste tipping fee, total fixed capital investment cost, diesel consumption, residue disposal cost, and electricity consumption. Due to the high variability of waste composition and PCRM market prices, the study evaluates the NPV's sensitivity to variation in waste composition and recovered materials selling prices. The critical parameters varied for LCA impact categories include diesel, electricity, baling wire, transportation, and waste composition. Table S5 in SI shows the sensitivity analysis variables for the base, optimistic, and lower cases.

For the uncertainty analysis, we evaluate the effect given that the uncertainty of crucial parameters varied in the sensitivity analysis and waste composition on NPV and GWP. This study utilizes Monte Carlo simulation to generate 10,000 distinct combinations of waste composition and key parameters in the TEA and LCA. The analysis uses a triangular distribution to generate random combinations using the U.S. EPA waste stream and baseline parameters as the mean value. The triangular distribution was chosen because it is common in Monte Carlo analysis when the actual distribution is unknown (Nanthakumar et al., 2009).

3. Results and discussion

3.1. Techno-economic analysis

The cost for a 120,000 t/y single-stream MRF is estimated to be \$23.13 MM. This capital cost is in the range of \$22.4 MM – \$29.4 MM reported in the literature (Cimpan et al., 2016; Logan City Council, 2018; Pressley et al., 2015). An MRF feasibility report carried out by the Logan city council, Iowa, estimated the capital cost of a 60,000 t MRF which services about 93,000 households as \$29.4 MM (Logan City Council, 2018), while Cimpan et al. (2016) estimated the capital cost of building an MRF could range from \$7.42 MM for a 25,000 t/y facility to \$22.4 MM for a 100,000 t/y MRF (Cimpan et al., 2016). The capital cost recorded in this study is similar to the values recorded by Cimpan et al. (2016). The slight difference could be attributed to the difference in facility size. Logan city MRF and the MRF modeled in this study have similar operating capacities per hour, the main difference being that the Logan city facility only operates one shift daily. However, the significant \$6 MM difference in the capital cost is found to be attributed to the disparity in the equipment cost reported in this study and the Logan MRF report (Logan City Council, 2018). The Logan MRF report did not break down the equipment cost to allow for further investigation.

The total equipment investment cost accounted for about 20% of the TFCI. The optical sorters for glass (19%), HDPE (10%), and PET (5%) account for about 34% of the total equipment investment cost. The one-way and two-way balers also contribute about 23% of the total equipment investment cost. This study assumes a fixed MRF configuration for all waste streams considered. Therefore, the waste composition does not affect the capital cost distribution.

The fixed operating cost for the 120,000 t/y facility is 45.48 \$/t of waste processed. The variable operating cost varies with the processed waste stream. Due to some of the assumptions in this study, the diesel price, waste disposal, transportation, and collection cost vary with the quantity of waste processed in the MRF. However, the baling wire and electricity requirements vary with the waste composition. Section 3.1.1 in the SI gives more information on the baling wire and electricity requirements for all waste streams considered in the study. Recovered aluminum is this study's most valuable recoverable material at \$1,708 /t, followed by HDPE and plastic film (Figure S1). Fig. 4b shows the total revenue generated from each waste stream and how much each recycled material contributes to the total revenue. The U.S. EPA generated the largest annual revenue because of its high HDPE, plastic films, and aluminum content. Seattle showed the lowest revenue due to its low aluminum and high OCC and Non-OCC fiber content.

The operating cost, capital cost, and revenue were used in a DCFROR analysis to estimate the NPV of the facility across the 20 years lifetime. Fig. 4a shows the NPV for the four waste streams. The NPV recorded for the U.S. EPA average waste stream showed a net present value of \$59 MM. The TEA model resulted in an NPV value of \$3.57 MM, \$30.54 MM, and \$43.39 MM for Seattle, Oregon, and Pennsylvania's waste compositions. The difference in NPV is because of the revenue generated from the different waste streams. Fig. 4b shows that the U.S. EPA, Oregon, and Pennsylvania composition generates more revenue than the Seattle waste stream. Seattle generates the lowest revenue because the waste composition has high glass, OCC, and non-OCC fibers, which sell for lower prices. Whereas the other waste streams have more valuable materials like aluminum and plastics. Glass increases the operating cost because of the high electricity required for sorting glass but contributes less to the revenue, hence, reducing the NPV of the system. Plastics improve the MRF's economic viability but are a small component of the waste streams recovered in the MRFs. However, the NPV (Fig. 4a) showed that the MRF is profitable for the four waste streams considered in this study.

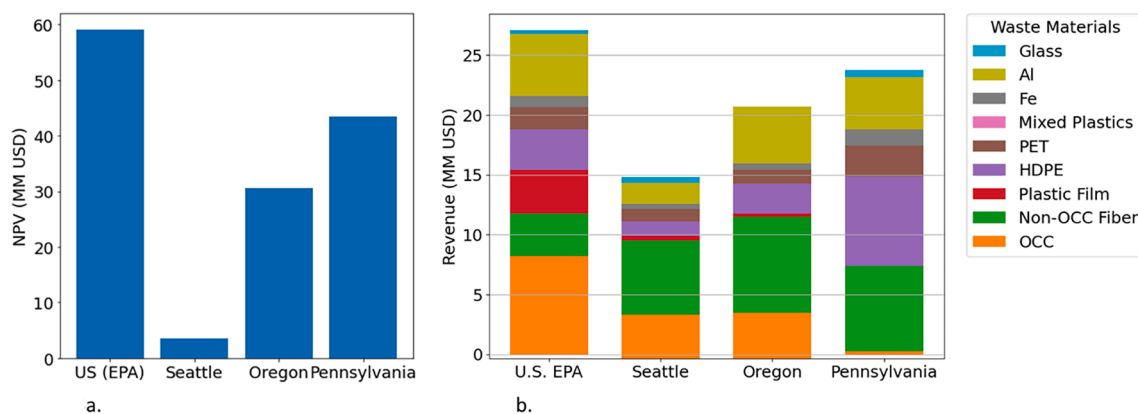


Fig. 4. A. Net Present Value (NPV) of the material recovery facility processing post-consumer recovered materials for the U.S. EPA, Seattle, Oregon, and Pennsylvania waste composition. Fig. 4b. Revenue from sales of recyclables recovered from U.S. EPA, Seattle, Oregon, and Pennsylvania's waste stream. (OCC – Old corrugated cardboard, HDPE – High-density polyethylene, Fe – Ferrous metals, Al – Aluminum).

3.2. Life cycle assessment

The LCA shows that the 100-year global warming potential (GWP) for processing the various waste streams ranges from 5.98 to 8.53 kg CO₂-eq /t of waste, with the Pennsylvania waste stream yielding the highest and Oregon the lowest emissions. Processing the U.S. EPA waste stream through the MRF resulted in 7.20 kg CO₂-eq /t of waste emissions. If the electricity mix of the states where the waste composition was collected is considered, the analysis resulted in GWP of 4.2, 6.7, and 4.3 kg CO₂-eq /t of waste for Seattle, Pennsylvania, and Oregon, respectively. These values are smaller than those recorded when the waste streams' GWP was evaluated using the average U.S. electricity mix. For Oregon and Seattle, hydroelectricity forms a significant percentage of the energy sources, reducing its GWP emission factors to 0.15 and 0.10 kg CO₂-eq / kWh, respectively. The range of emissions reported in this study is lower when compared to landfilling emissions of 15.02 kg CO₂-eq/t of mixed plastic waste and 1552.6 kg CO₂-eq/t of mixed paper waste reported in (Demetriou & Crossin, 2019).

The range of the GWP recorded both by using the average U.S. electricity mix, and individual state's electricity mix is similar to the one recorded by Smith et al., (2022) where they reported a GWP of 8.58 kg CO₂-eq/t of waste. They used average values from various studies for electricity, diesel, and baling wire requirements to arrive at the emission

value (Smith et al., 2022). Fitzgerald et al. (2012) reported emissions of 11.07 kg CO₂-eq /t of waste processed. The difference in these reported emissions and those of Fitzgerald et al. (2012) can be attributed to assuming that more energy-efficient lighting and equipment are employed in the facilities. In rare scenarios where the PCRM from an MRF can replace virgin materials in the economy without secondary processing, the potential environmental savings are shown in Fig. 5 and Table S10, with GWP ranging from 1388 to 1809 kg CO₂ eq. However, when secondary processing of the PCRM is considered, the savings ranged from –656 kg CO₂ eq for U.S. EPA stream to –1,040 kg CO₂ eq for the Pennsylvania waste stream. Compared to emissions reported for alternative waste management strategies such as waste-to-energy (–36 to 40 kg CO₂-eq /M.T. of waste), recovering waste in the MRF is a more environmentally friendly approach from a GWP perspective. In-depth analysis of secondary processing is beyond the scope of this study but discussions on substitution ratios and GWP data can be found in section 3.2 in SI. Also, information on the contributions of electricity, diesel, baling wire, and collection and transportation to all impact categories can be found in section 3.2 in SI.

3.3. Sensitivity and uncertainty analysis

Fig. 6 shows the sensitivity analysis results of NPV for processing the

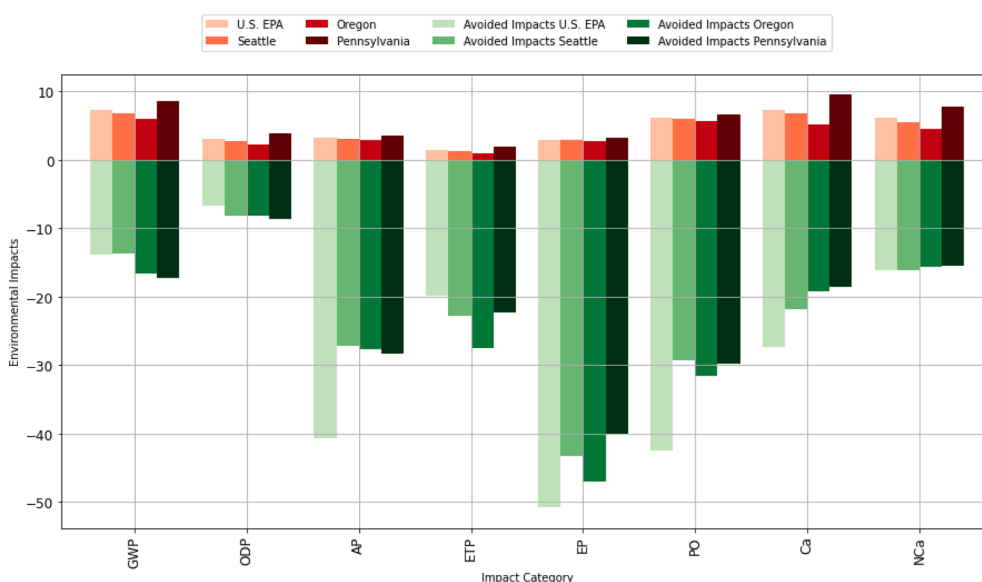


Fig. 5. Health and environmental impacts and savings for the U.S. EPA, Seattle, Pennsylvania, and Oregon scenarios. The red bars represent impacts from recovering the materials, while the green bars represent savings by replacing virgin. GWP (kg CO₂-eq), AP (H + -eq), EP (kg N), ETP (kg 2,4-D-eq), ODP (CFC-11-eq), PO (kg NO_x-eq), Ca (kg benzene-eq) and Nca (kg toluene-eq), Avoided GWP (10E-2 kg CO₂-eq), Avoided AP (10E-1H + -eq), Avoided EP (10 kg N), Avoided ETP (10E-2 kg 2,4-D-eq), Avoided ODP (10E5 CFC-11-eq), Avoided PO (10 kg NO_x-eq), Avoided Ca (10 kg benzene-eq) and Avoided Nca (10E-3 kg toluene-eq); (GWP- Global warming potential, ODP- Ozone depletion, AP - Acidification, ETP- Ecotoxicity, EP - Eutrophication, PO Photochemical oxidation, Ca - Carcinogenic, NCa - Non-carcinogenic). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

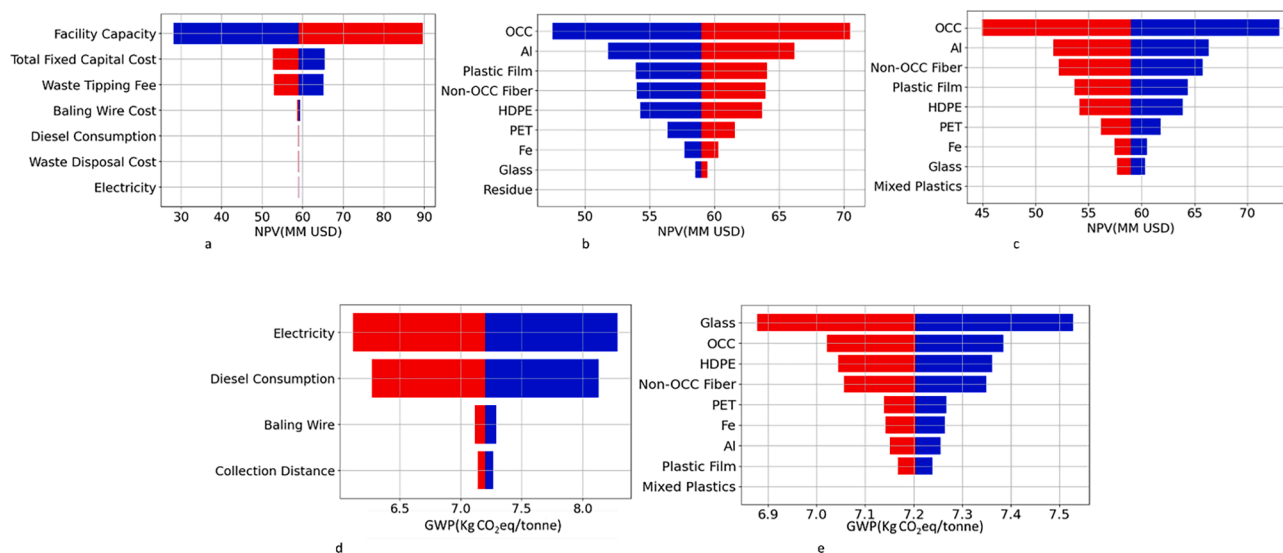


Fig. 6. Sensitivity analysis of net present value (NPV) and global warming potential (GWP) for the U.S. EPA scenario. **Fig. 6a.** Sensitivity of NPV to key facility parameters. **Fig. 6b.** Sensitivity of NPV to post consumer recovered materials prices. **Fig. 6c.** Sensitivity of NPV to waste composition. **Fig. 6d.** Sensitivity of GWP to key parameters. **Fig. 6e.** Sensitivity of GWP to waste composition.

U.S. EPA waste stream to key parameters, waste composition, and post-consumer recovered material prices. The plant capacity, total fixed capital cost, and waste tipping fee had the most impact on the NPV of the MRF. A 30% increase in the plant capacity can increase the NPV from \$58.95 MM to \$90 MM, while a 30% decrease in plant capacity can result in an NPV lower than \$30 MM. Baling wire, diesel, waste disposal, and electricity costs show little effect on the NPV. Figure S6 in the S.I. shows the sensitivity analysis results for Oregon, Seattle, and Pennsylvania's waste streams to key parameters. The figures show similar sensitivity results to the U.S. EPA waste composition. However, the NPV in Seattle's waste stream resulted in a negative NPV with a 30% reduction in the most impactful parameters. This is due to the revenue generated from the Seattle waste stream being smaller than the other waste streams. Other parameters that can largely influence the NPV include total fixed capital cost and waste tipping fees. A $\pm 30\%$ change in the selling prices of OCC and Non-OCC fiber can result in a large change in the NPV in the U.S. EPA waste stream scenario because OCC and Non-OCC represent about 70% of the total waste composition. Aluminum represents about 2.6% of the U.S. EPA waste, but due to its high selling price, the NPV is very sensitive to variation in its composition and prices. Figure S7 and figure S8 in SI show the NPV sensitivity to waste composition and market price analysis results for the other waste streams considered in this study. The results show similar trends, with materials dominating the waste composition and materials with high selling prices significantly affecting the NPV.

Figure S9 to Figure S12 in the SI show the LCA sensitivity analysis results for all impact categories considered in this study for the U.S. average, Seattle, Oregon, and Pennsylvania's waste streams. The sensitivity results show that diesel and electricity consumption significantly impact the environmental burdens of processing recyclables at the MRF. Figure S9 shows the impact of diesel consumption, electricity consumption, baling wire, and collection distance on the GWP associated with recovering valuable recyclables from the U.S. EPA waste stream. A $\pm 30\%$ change in diesel or electricity consumption can increase or decrease the GWP from 7.20 kg CO₂-eq to 8.29 kg CO₂-eq and 6.12 kg CO₂-eq, a 15% change in the baseline value. Baling wire and collection distance do not significantly impact GWP. When analyzing the individual waste streams at the MRF for the U.S. EPA composition, glass shows the most significant impact on GWP. A $\pm 30\%$ change in the quantity of glass in the waste stream can result in about a 5% change in the GWP (7.20 to 7.5 or 6.9 kg CO₂-eq / t). This is not surprising as electricity contributes

significantly to GWP, and glass-recovering equipment is energy intensive. A $\pm 30\%$ variation in the amount of PET, Fe, Al, plastic film, and mixed plastics in the waste stream has a minimal effect; less than 1% change on the GWP. This is expected as these materials represent less than fifteen percent of the total composition. Figure S13 to Figure S16 in SI show the sensitivity to waste composition analysis results for the four waste compositions considered in this study. Results and discussion on uncertainty analysis can be found in section 3.3.2 in SI.

4. Conclusions

This study investigated the economic feasibility and environmental impacts of a 120,000 t/y commercial-scale standalone material recovery facility. The study estimated the NPV of the facility using the discounted cash flow rate of return analysis. The facility's capital cost is \$23 MM, and the operating cost for processing U.S. EPA waste composition is 45.48 \$/t. The NPV for the baseline composition U.S. EPA is \$58.97. The analysis shows that the NPV is very sensitive to waste composition, as evidenced in the NPV of \$3.75 MM, \$30.54 MM, and \$43.99 MM for Seattle, Oregon, and Pennsylvania waste compositions. It can be inferred that waste stream containing significant fractions of aluminum, HDPE, plastic film, and PET will result in a more substantial NPV due to their high selling prices. Sensitivity analysis shows that facility capacity, total fixed capital cost, and waste tipping fees are important drivers of the NPV. A 30% change in any waste tipping fees or total fixed capital cost significantly influences the NPV.

The LCA estimated the health and environmental burdens associated with processing 1 t of recyclable waste the MRF. The LCA results show the 100-year GWP of processing the U.S. EPA waste stream to be 7.20 kg CO₂-eq / t. The GWP also varied with the composition of the waste. Pennsylvania's waste stream had the most significant emissions of 8.53 kg CO₂-eq / t. Sensitivity analysis shows that electricity and diesel are substantial drivers of 100-year global warming potential, acidification potential, eutrophication potential, ecotoxicity, ozone depletion, and photochemical oxidation. LCA results also show that a waste stream with a significant amount of materials recovered with energy-intensive equipment results in more substantial health and environmental burdens.

These results indicate that the composition of waste streams can significantly impact the profitability and environmental impacts of an MRF. The MRF design could be optimized to reduce costs and emissions

by focusing on select products. Future work would investigate configuration and operational changes that MRF facilities could take to optimize the economics and sustainability of their operations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasman.2023.05.011>.

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